Chapter Notes: Changes Around Us - Physical and Chemical

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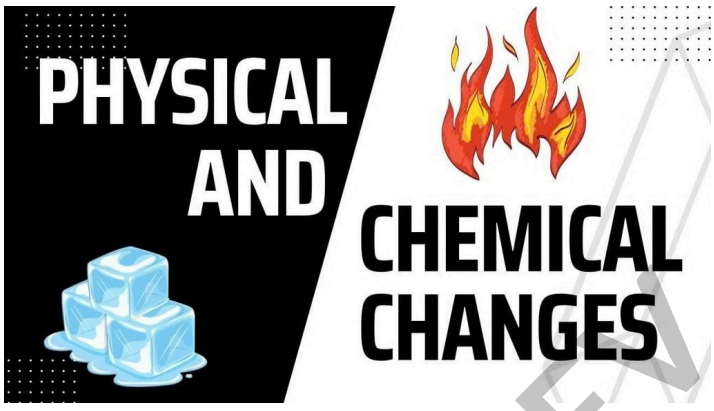
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Introduction

We observe many changes around us every day. Ice melts into water, flowers open from buds, fruits change colour and smell, and cold water becomes warm over time. These changes alter how things look, smell, feel, sound or taste. By using our five senses, namely **sight**, **smell**, **touch**, **hearing** and **taste**, we notice and describe these changes.



In this chapter, we learn to distinguish between two broad kinds of changes: **physical changes** and **chemical changes**. We also study everyday examples, simple tests for detecting some chemical changes, and why some changes are reversible while others are not.

Physical Change and Chemical Change

Physical Change

Definition: A physical change is a change in which the appearance, shape, size or state of a substance changes, but its chemical composition remains the same. No new substance is formed.



Physical Changes

Key features of physical changes

Only physical properties such as shape, size or state change.
The chemical composition of the substance remains unchanged.
Physical changes are often reversible by simple physical means, although some physical changes are not easily reversible.
No new substance with new chemical properties is formed.

Examples of physical changes

Paper folding: Folding paper changes its form. Unfolding it can restore the sheet to its original shape.



Balloon inflation: Inflating a balloon stretches it. Letting the air out returns it towards its original shape. If the balloon bursts, the rubber is still the same material, but its original shape cannot be restored.

Crushing chalk: Crushing chalk into powder changes its form but not its chemical identity. The original piece usually cannot be reformed by simple methods.

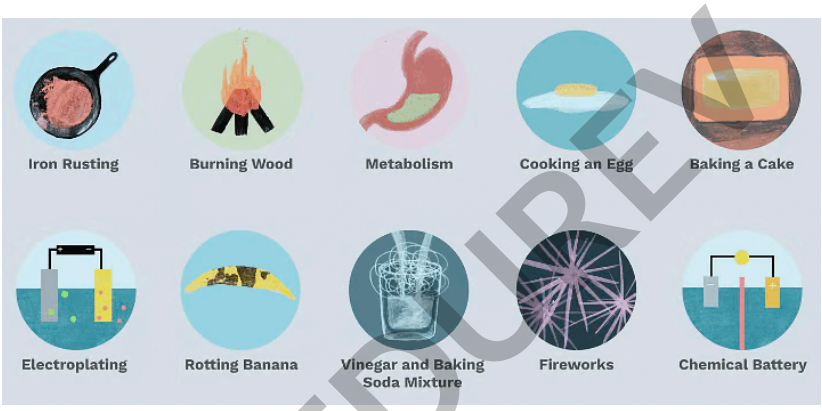
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Change of states of water: Ice changes into water and water changes into steam. These are physical changes because the substance remains water throughout.

Chemical Change

Definition: A chemical change, or chemical reaction, is a process in which one or more new substances with different properties are formed. Chemical changes are usually not easily reversed by simple physical methods.



Chemical Changes

A chemical change can often be recognised by one or more of the following observations:

Formation of a new substance.

Evolution of a gas.

Formation of a precipitate, which is an insoluble solid formed in a solution.

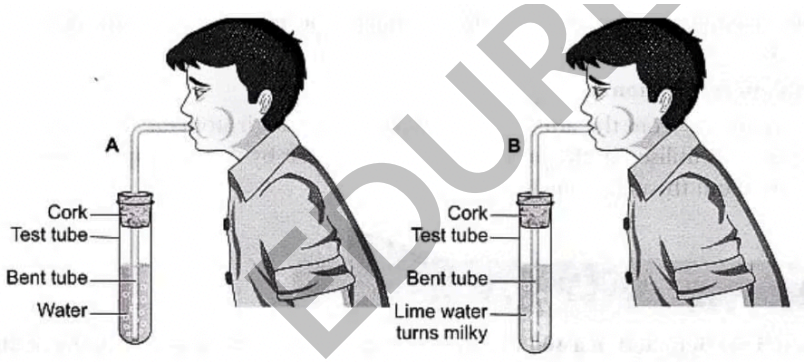
Change in colour or smell.

Production or absorption of heat and sometimes light.

Examples and Explanations

1. Blowing Air into Lime Water

Air from our breath contains **carbon dioxide**. When it is bubbled through **lime water**, which is a solution of calcium hydroxide, a chemical reaction occurs.



The lime water turns milky because a white, insoluble solid called **calcium carbonate** is formed. This white material later settles as a precipitate. Water is also produced.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

This experiment shows that a new substance has been produced. Therefore, it is a chemical change.

2. Vinegar and Baking Soda

When **vinegar**, which contains acetic acid, is mixed with **baking soda**, which is sodium bicarbonate, a vigorous chemical reaction occurs.



The reaction produces bubbles or fizzing because **carbon dioxide gas** is released. The gas turns lime water milky when it is bubbled through it. This confirms that the gas is carbon dioxide.

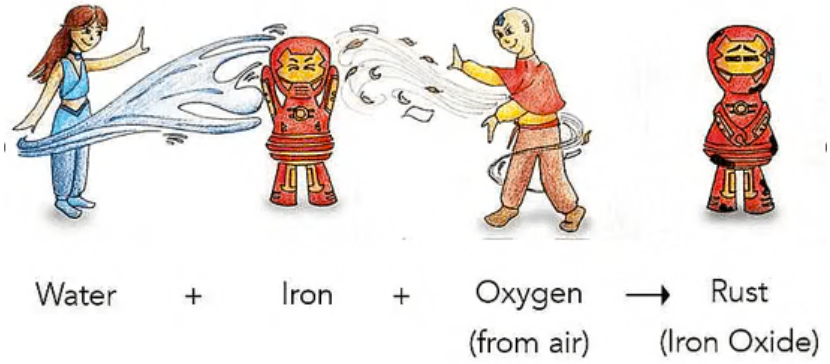
$$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + CO_2 + H_2O$$

The bubbling is direct evidence of gas formation.
New chemical substances, such as sodium acetate, are also produced.

Some Other Processes Involving Chemical Changes

Rusting

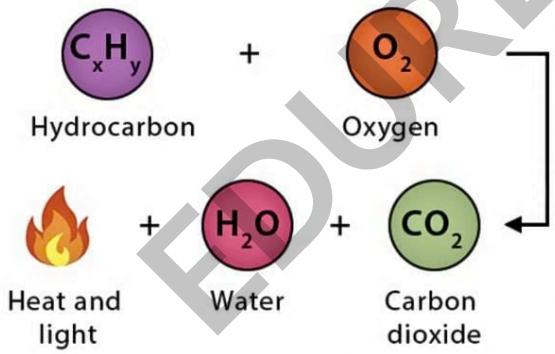
Rusting is a chemical change in which iron reacts with oxygen in the presence of moisture to form rust. Rust is a new substance and is generally brown and flaky. Rust weakens iron objects such as nails, tools and bridges because it has different mechanical and chemical properties from iron.



Rusting of Iron

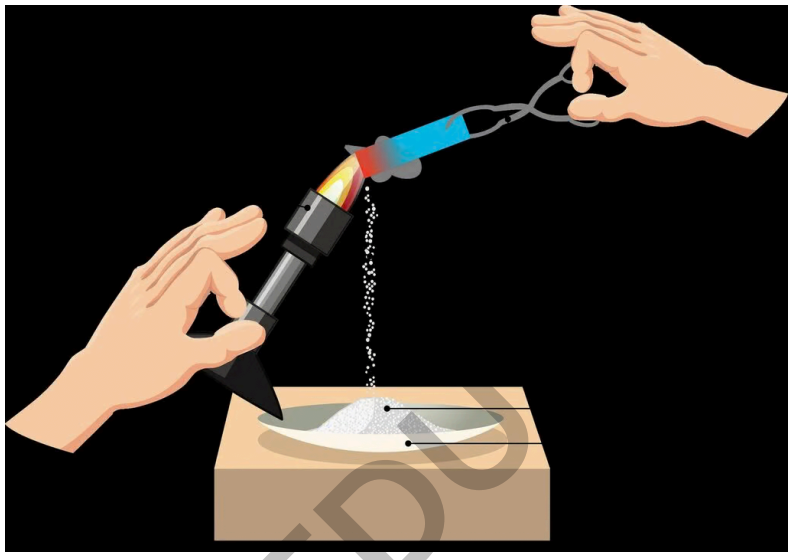
Combustion

Combustion is a chemical reaction in which a substance reacts with oxygen to produce heat and often light. Materials that burn are called **combustible substances**. Examples include wood, paper, cotton and kerosene.



Burning a magnesium ribbon in air produces a white solid called **magnesium oxide**. This shows that a new substance is produced during combustion.
Combustion reactions release energy in the form of heat and sometimes light.

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The balanced chemical equation for burning magnesium is:

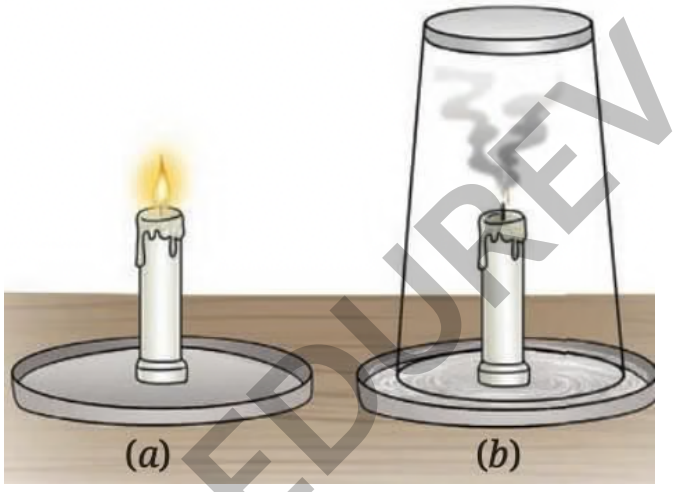
$$2Mg + O_2 \rightarrow 2MgO$$

Oxygen's Role in Combustion

Oxygen from the air is essential for most combustion reactions.

If a burning candle or another fire is covered so that it cannot receive fresh air, the flame goes out because the available oxygen is gradually used up.

During the combustion of a fuel containing carbon, carbon reacts with oxygen to form **carbon dioxide**. This gas can be detected by passing it through lime water, which turns milky.



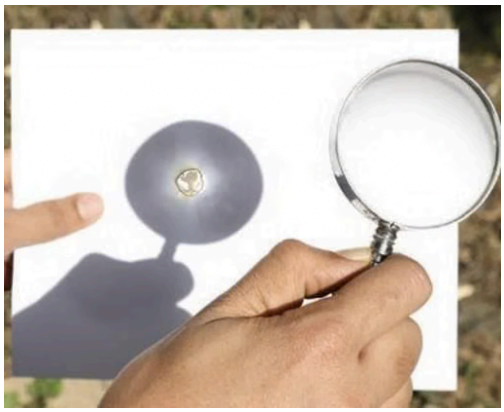
Candle (a) burning (b) covered with a glass tumbler

What Else Is Needed to Start Combustion?

Besides a **fuel** and **oxygen**, **heat** is needed to start combustion. The minimum temperature at which a substance starts burning is called its **ignition temperature**.



A lighted match supplies the heat needed to raise paper to its ignition temperature, after which the paper burns.
Concentrated sunlight, using a magnifying glass, can also heat paper to its ignition temperature and set it on fire.



Paper catching fire

Fascinating Fact

Have you seen tiny glowing insects in gardens at night? They are called **fireflies**. Their light is produced by a chemical reaction inside their bodies and is an example of **bioluminescence**. This glow is produced with very little heat.



Science and Society: Safety

What should be done if someone's clothes catch fire?

Wrap the person in a **blanket or cloth** to cut off the air supply and stop the flames. The person should not run, because running can make the fire burn more strongly. If possible, use thick, non-synthetic material.





Important: Do not use synthetic blankets or clothing because they may melt and stick to the skin, causing more severe injuries.

Can Physical and Chemical Changes Occur in the Same Process?

1. Burning a Candle: Physical Changes

Melting of wax: Heat from the flame melts solid wax into liquid wax. This is a physical change because the chemical identity of the wax does not change.

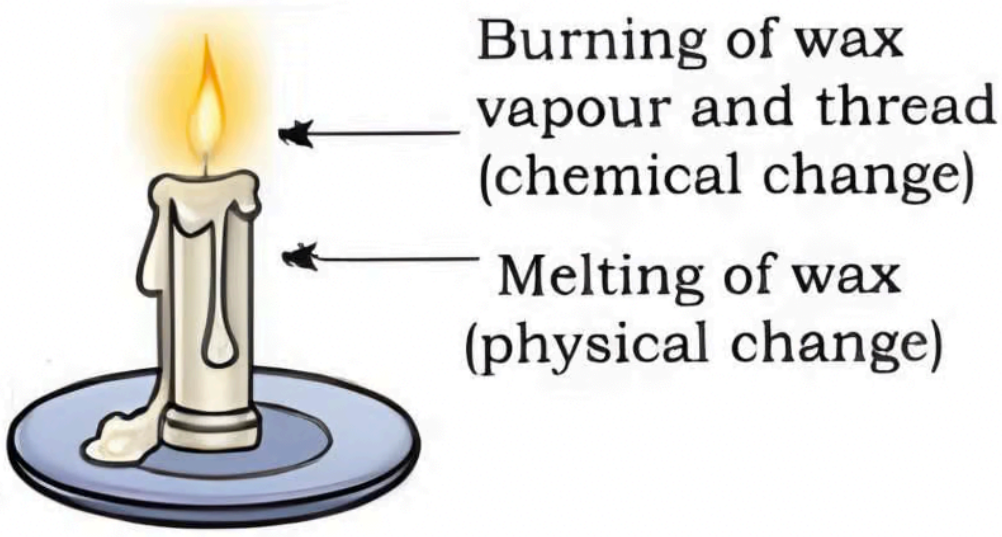
Vapourisation: Liquid wax near the wick changes into vapour and travels upwards through the wick. This is a physical change.

Solidification: Wax that drips and cools changes back into a solid. This is another physical change.

2. Burning a Candle: Chemical Change

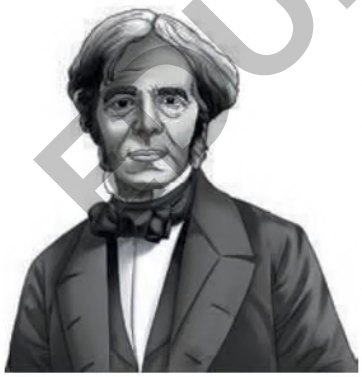
Combustion of wax vapour: Wax vapour burns in oxygen to produce **carbon dioxide**, **water**, heat and light. New substances are formed, so this part is a chemical change.

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Conclusion: Burning a candle involves both physical changes, such as melting, vapourisation and solidification of wax, and a chemical change, namely the combustion of wax vapour to form new substances.

Know a Scientist: Michael Faraday



Michael Faraday, a nineteenth-century scientist, used candles to explain physical and chemical processes in simple language. His popular lectures titled **"Chemical History of a Candle"** describe melting, vapourisation and combustion clearly for learners.

Are Changes Permanent?

Reversible Changes

Definition: Reversible changes are changes in which the original substance or object can be restored by simple physical methods.



Examples:

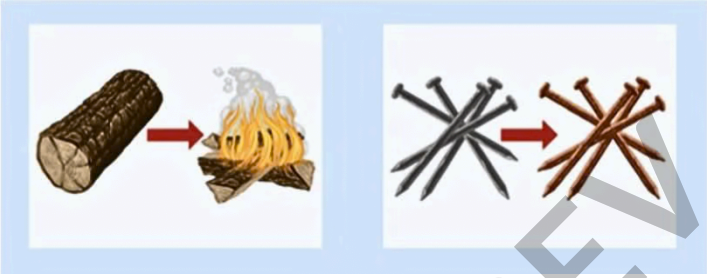
Melting ice: Ice melts into water, and freezing the water reforms ice.

Boiling water: Water changes into steam, and condensing steam returns it to liquid water.

Folding paper: Folding and then unfolding paper can restore its original shape.

Irreversible Changes

Definition: Irreversible changes are changes in which the original substance or object cannot be restored by simple physical means.



Examples:

Chopping vegetables: Cut pieces cannot be joined back to form the whole vegetable.

Making popcorn: Corn kernels transform into popcorn and cannot change back into kernels.

Burning wood: Wood changes into ash, and ash cannot become wood again.

Are All Changes Desirable?

Desirable Changes

Desirable changes are useful or beneficial to us.

Examples:

Milk to curd: Milk ferments to form curd, which is edible and nutritious.

Ripening fruits: Fruits change colour, become sweeter and become fit to eat.

Cooking food: Raw ingredients change into cooked food, which is easier to digest and safer to eat.

Cutting fruits: Cutting fruits makes them ready to eat.

Undesirable Changes

Undesirable changes are harmful or unwanted.

Examples:

- **Rusting of iron:** Rusting damages tools and structures.
- **Food decay:** Decay spoils stored food and makes it unfit to eat.

Some changes may be useful in one situation and harmful in another. For example, decomposition of kitchen waste is undesirable when it causes a foul smell and disease, but the same process is useful for making compost for plants.

Environmental Impact

Human activities such as burning fossil fuels in vehicles and industries release **carbon dioxide** and other pollutants into the air.

These emissions increase atmospheric pollution and contribute to long-term changes such as **climate change**.



Some Slow Natural Changes

Weathering of Rocks

Weathering is the process by which rocks break down into smaller pieces called sediments through physical and chemical processes. Over long periods, these sediments contribute to soil formation.



Types of weathering

Physical weathering: This is caused by temperature changes, expansion and contraction, plant roots growing into cracks, or water freezing and expanding in crevices. These actions break rocks into smaller pieces such as sand and gravel.

Chemical weathering: This occurs when water or dissolved chemicals react with minerals in rocks and change their composition. For example, iron-containing rocks may form a reddish layer of iron oxide, similar to rust, when exposed to water and air.

Both types of weathering, occurring over long periods, help form soil, which is essential for plant growth and life on Earth.

Erosion

Erosion is the movement or removal of weathered rock, soil and sediments by natural forces such as flowing water, wind or glaciers.



Examples and effects

Sand on riverbeds: Large rocks and stones are worn down and moved by rivers, producing fine sand and pebbles.

Smoothing of river rocks: Flowing water rubs stones against one another, making them smooth over time.

Landslides: Rapid erosion during landslides moves large amounts of soil and rock downhill. This is a fast physical change.

When wind or water slows down, the carried sediments settle. Over thousands of years, they can be compacted and hardened into new rocks.

These changes are usually slow and often irreversible on human timescales.

Terms to Remember

Physical changes: Only shape, size or state changes; no new substance is formed. Examples include melting ice, boiling water and folding clothes.

Chemical changes: New substances are formed. Examples include rusting, burning and lime water turning milky when carbon dioxide is passed through it.

Rusting: Iron reacts with air and water to form rust, a form of iron oxide. It is a chemical change.

Combustion: Burning in oxygen produces heat and often light. It requires fuel, oxygen and sufficient heat to reach the ignition temperature.

Fireflies: Fireflies glow because of a chemical reaction called **bioluminescence**, which produces light with very little heat.

Burning a candle: It involves physical changes, such as wax melting and vapourisation, and a chemical change, in which vapour burns to form carbon dioxide and water.

Reversible changes: Changes that can be reversed by simple means, such as melting and freezing of water.

Irreversible changes: Changes that cannot be easily reversed, such as cutting and burning.

Desirable changes: Useful or beneficial changes, such as cooking, making curd and ripening fruits.

Undesirable changes: Harmful or unwanted changes, such as rusting and food spoilage.

Weathering: The breakdown of rocks into smaller pieces and soil by physical and chemical processes.

Erosion: The movement of soil and sediments by wind, water or ice, which shapes the land.

Michael Faraday: A scientist who explained physical and chemical processes using everyday examples such as candles in his lectures **"Chemical History of a Candle"**.

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You can practise questions from this chapter here: **HOTS Questions: Changes Around Us: Physical and Chemical** (https://edurev.in/studytube/Class-7-Science-Chapter-5-HOTS-Questions--Changes-Around-Us-Physical-and-Chemical/87c22e94-f79c-4b9d-a748-63a5806c5e26_t)